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Section: BCS-243D

Syllabus: Database Systems Lab

Syllabus Code: CL2005

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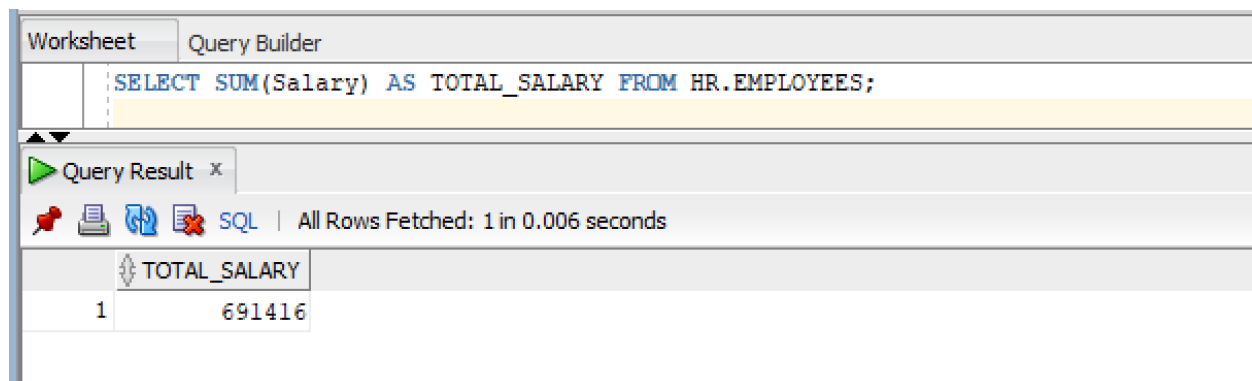
Q1)

1)

SQL Query:

```
SELECT SUM(Salary) AS TOTAL_SALARY FROM HR.EMPLOYEES;
```

Snapshot of Query:



The screenshot shows a database query tool interface. At the top, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying the SQL query: `SELECT SUM(Salary) AS TOTAL_SALARY FROM HR.EMPLOYEES;`. Below the query, there is a 'Query Result' section. It shows a table with one column, 'TOTAL_SALARY', and one row with the value '691416'. The status bar indicates 'All Rows Fetched: 1 in 0.006 seconds'.

	TOTAL_SALARY
1	691416

2)

SQL Query:

```
SELECT ROUND(AVG(Salary), 2) AS AVERAGE_SALARY FROM HR.EMPLOYEES;
```

Snapshot of Query:

Worksheet

Query Builder

```
SELECT ROUND(AVG(Salary), 2) AS AVERAGE_SALARY FROM HR.EMPLOYEES;
```

▶ Query Result x

 SQL | All Rows Fetched: 1 in 0.004 seconds

	AVERAGE_SALARY
1	6461.83

3)

MAX

SQL Query:

```
SELECT MAX(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;
```

Snapshot of Query:

Worksheet

Query Builder

```
SELECT MAX(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;
```

 Query Result x

 SQL | All Rows Fetched: 1 in 0.004 seconds

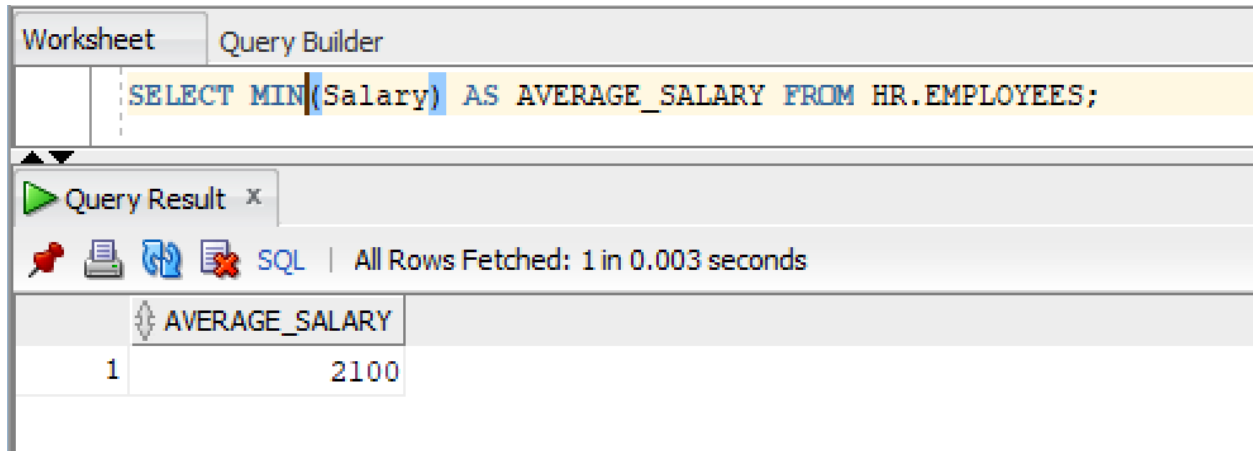
	AVERAGE_SALARY
1	24000

MIN

SQL Query:

SELECT MIN(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;

Snapshot of Query:



The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the query: `SELECT MIN(Salary) AS AVERAGE_SALARY FROM HR.EMPLOYEES;`. Below the query, the 'Query Result' tab shows the execution status: 'All Rows Fetched: 1 in 0.003 seconds'. The result is displayed in a table with one column, 'AVERAGE_SALARY', and one row with the value '2100'.

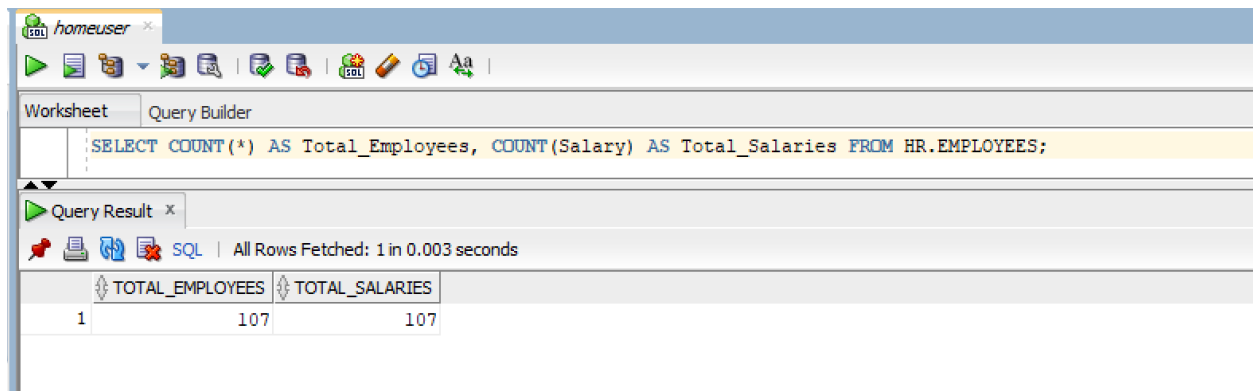
	AVERAGE_SALARY
1	2100

4)

SQL Query:

SELECT COUNT(*) AS Total_Employees, COUNT(Salary) AS Total_Salaries FROM HR.EMPLOYEES;

Snapshot of Query:



The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the query: `SELECT COUNT(*) AS Total_Employees, COUNT(Salary) AS Total_Salaries FROM HR.EMPLOYEES;`. Below the query, the 'Query Result' tab shows the execution status: 'All Rows Fetched: 1 in 0.003 seconds'. The result is displayed in a table with two columns, 'TOTAL_EMPLOYEES' and 'TOTAL_SALARIES', and one row with the values '107' and '107'.

	TOTAL_EMPLOYEES	TOTAL_SALARIES
1	107	107

5)

SQL Query:

```
SELECT COUNT(DISTINCT DEPARTMENT_ID) AS Unique_Departments FROM  
HR.EMPLOYEES;
```

Snapshot of Query:

The screenshot shows a SQL query editor with a 'Query Builder' tab. The query text is: `SELECT COUNT(DISTINCT DEPARTMENT_ID) AS Unique_Departments FROM HR.EMPLOYEES;`. Below the query editor is a 'Query Result' window. It displays a table with one column named 'UNIQUE_DEPARTMENTS' and one row with the value '11'. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

UNIQUE_DEPARTMENTS
11

Q2)

1)

SQL Query:

```
SELECT DEPARTMENT_ID, SUM(SALARY) AS Dept_Total_Sal
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID;
```

Snapshot of Query:

WorksheetQuery Builder

```
SELECT DEPARTMENT_ID, SUM(SALARY) AS Dept_Total_Sal
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID;
```

Query Result x

 SQL | All Rows Fetched: 12 in 0.003 seconds

	DEPARTMENT_ID	DEPT_TOTAL_SAL
1	100	51608
2	30	24900
3	(null)	7000
4	90	58000
5	20	19000
6	70	10000
7	110	20308
8	50	156400
9	80	304500
10	40	6500
11	60	28800
12	10	4400

2)

SQL Query:

```
SELECT JOB_ID, ROUND(AVG(SALARY), 2) AS Job_Avg_Sal
FROM HR.EMPLOYEES
GROUP BY JOB_ID;
```

Snapshot of Query:

Worksheet

Query Builder

```
SELECT JOB_ID, ROUND(AVG(SALARY), 2) AS Job_Avg_Sal
FROM HR.EMPLOYEES
GROUP BY JOB_ID;
```

Query Result x

    SQL | All Rows Fetched: 19 in 0.004 seconds

	JOB_ID	JOB_AVG_SAL
1	IT_PROG	5760
2	AC_MGR	12008
3	AC_ACCOUNT	8300
4	ST_MAN	7280
5	PU_MAN	11000
6	AD_ASST	4400
7	AD_VP	17000
8	SH_CLERK	3215
9	FI_ACCOUNT	7920
10	FI_MGR	12008
11	PU_CLERK	2780
12	SA_MAN	12200
13	MK_MAN	13000
14	PR_REP	10000
15	AD_PRES	24000
16	SA_REP	8350
17	MK_REP	6000
18	ST_CLERK	2785
19	HR_REP	6500

3)

SQL Query:

```
SELECT DEPARTMENT_ID, MAX(SALARY) AS Dept_Max_Sal  
FROM HR.EMPLOYEES  
GROUP BY DEPARTMENT_ID;
```

Snapshot of Query:

Worksheet

Query Builder

```
SELECT DEPARTMENT_ID, MAX(SALARY) AS Dept_Max_Sal
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID;
```

Query Result x

    SQL | All Rows Fetched: 12 in 0.004 seconds

	DEPARTMENT_ID	DEPT_MAX_SAL
1	100	12008
2	30	11000
3	(null)	7000
4	90	24000
5	20	13000
6	70	10000
7	110	12008
8	50	8200
9	80	14000
10	40	6500
11	60	9000
12	10	4400

4)

SQL Query:

```
SELECT DEPARTMENT_ID, JOB_ID, COUNT(*) AS EMPLOYEE_COUNT
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID, JOB_ID
ORDER BY DEPARTMENT_ID;
```

Snapshot of Query:

Worksheet

Query Builder

```
SELECT DEPARTMENT_ID, JOB_ID, COUNT(*) AS EMPLOYEE_COUNT
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID, JOB_ID
ORDER BY DEPARTMENT_ID;
```

Query Result x

SQL | All Rows Fetched: 20 in 0.005 seconds

	DEPARTMENT_ID	JOB_ID	EMPLOYEE_COUNT
1	10	AD_ASST	1
2	20	MK_MAN	1
3	20	MK_REP	1
4	30	PU_CLERK	5
5	30	PU_MAN	1
6	40	HR_REP	1
7	50	SH_CLERK	20
8	50	ST_CLERK	20
9	50	ST_MAN	5
10	60	IT_PROG	5
11	70	PR_REP	1
12	80	SA_MAN	5
13	80	SA_REP	29
14	90	AD_PRES	1
15	90	AD_VP	2
16	100	FI_ACCOUNT	5
17	100	FI_MGR	1
18	110	AC_ACCOUNT	1
19	110	AC_MGR	1
20	(null)	SA_REP	1

Q3)

1)

SQL Query:

```
SELECT DEPARTMENT_ID, COUNT(EMPLOYEE_ID) AS NUM_OF_EMP  
FROM HR.EMPLOYEES  
GROUP BY DEPARTMENT_ID  
HAVING COUNT(EMPLOYEE_ID) > 3;
```

Snapshot of Query:

Worksheet

Query Builder

SELECT DEPARTMENT_ID, COUNT(EMPLOYEE_ID) AS NUM_OF_EMP
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID
HAVING COUNT(EMPLOYEE_ID) > 3;

Query Result x

SQL | All Rows Fetched: 5 in 0.003 seconds

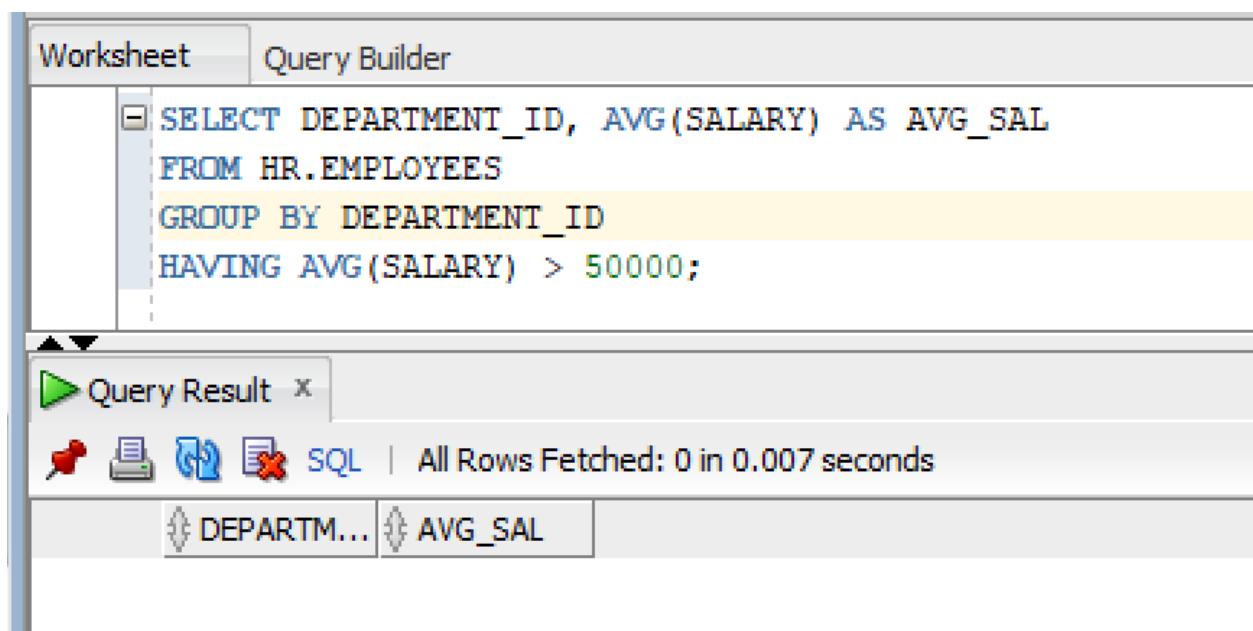
	DEPARTMENT_ID	NUM_OF_EMP
1	100	6
2	30	6
3	50	45
4	80	34
5	60	5

2)

SQL Query:

```
SELECT DEPARTMENT_ID, AVG(SALARY) AS AVG_SAL  
FROM HR.EMPLOYEES  
GROUP BY DEPARTMENT_ID  
HAVING AVG(SALARY) > 50000;
```

Snapshot of Query:



3)

SQL Query:

```
SELECT DEPARTMENT_ID, MIN(SALARY) AS MIN_SAL
FROM HR.EMPLOYEES
GROUP BY DEPARTMENT_ID
HAVING MIN(SALARY) < 25000;
```

Snapshot of Query:

Worksheet

Query Builder

SELECT DEPARTMENT_ID, MIN(SALARY) AS MIN_SAL

FROM HR.EMPLOYEES

GROUP BY DEPARTMENT_ID

HAVING MIN(SALARY) < 25000;

Query Result x

SQL | All Rows Fetched: 12 in 0.005 seconds

	DEPARTMENT_ID	MIN_SAL
1	100	6900
2	30	2500
3	(null)	7000
4	90	17000
5	20	6000
6	70	10000
7	110	8300
8	50	2100
9	80	6100
10	40	6500
11	60	4200
12	10	4400

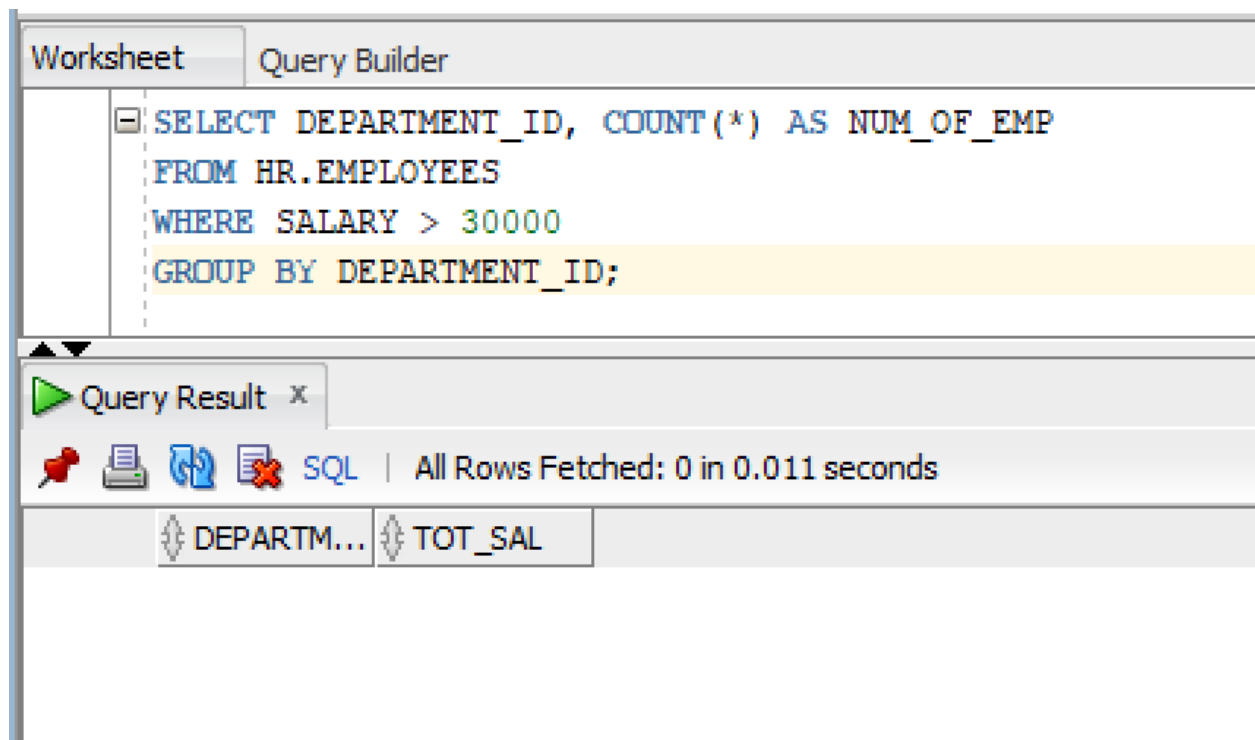
Q4)

1)

SQL Query:

```
SELECT DEPARTMENT_ID, COUNT(*) AS NUM_OF_EMP  
FROM HR.EMPLOYEES  
WHERE SALARY > 30000  
GROUP BY DEPARTMENT_ID;
```

Snapshot of Query:

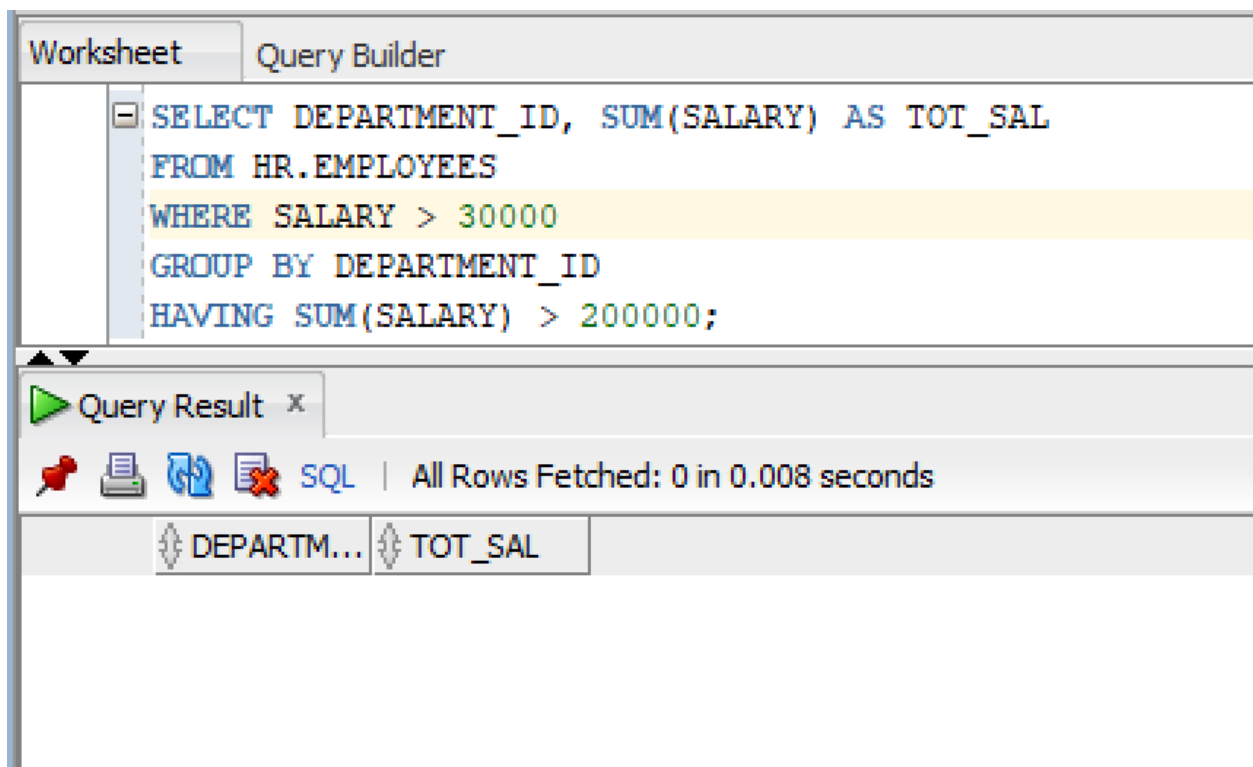


2)

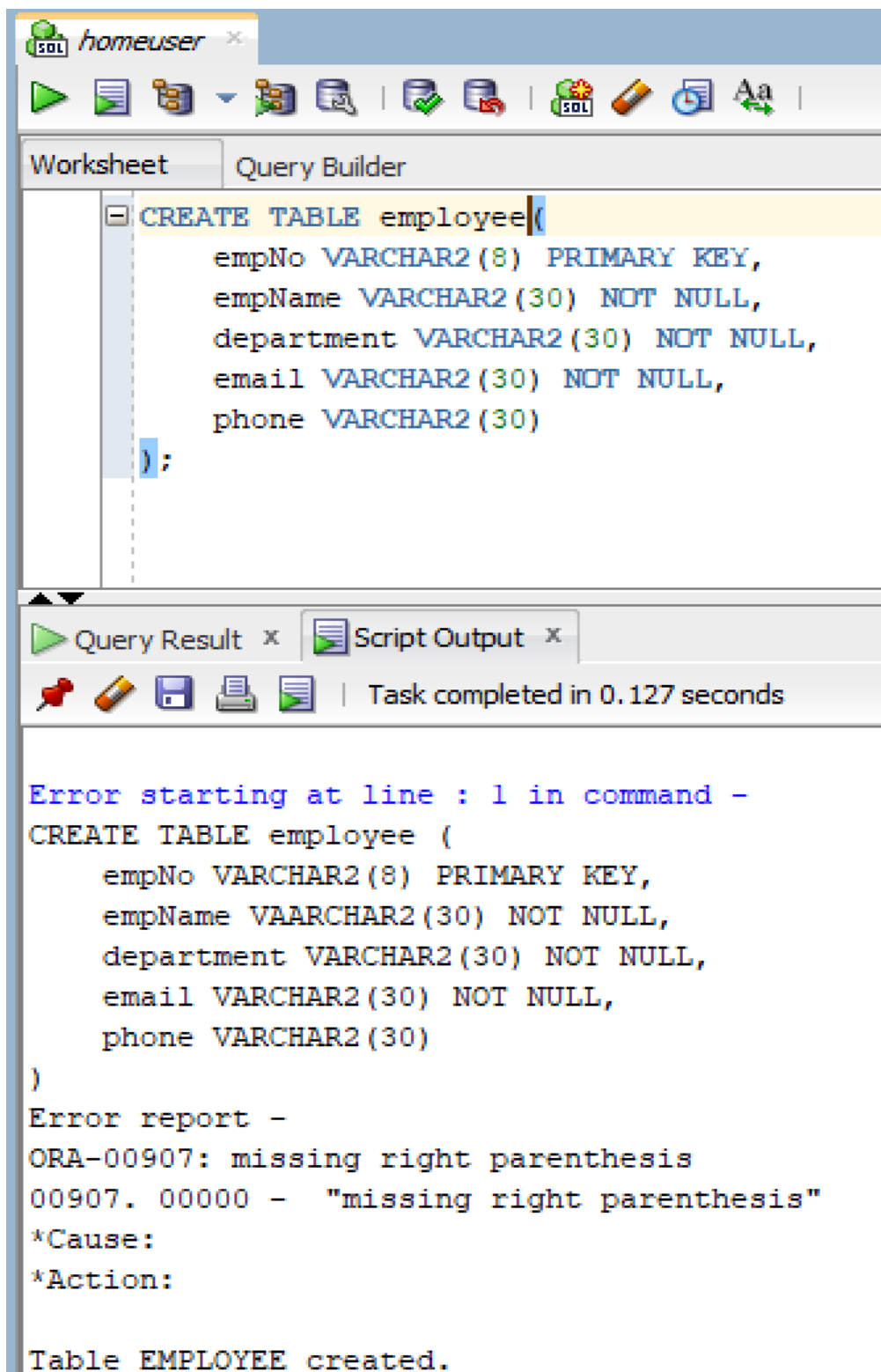
SQL Query:

```
SELECT DEPARTMENT_ID, SUM(SALARY) AS TOT_SAL  
FROM HR.EMPLOYEES  
WHERE SALARY > 30000  
GROUP BY DEPARTMENT_ID  
HAVING SUM(SALARY) > 200000;
```

Snapshot of Query:



PART B – DML OPERATIONS



The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and other database operations. The 'Query Builder' tab is active, displaying a SQL statement to create a table named 'employee'. The statement is as follows:

```
CREATE TABLE employee(  
    empNo VARCHAR2(8) PRIMARY KEY,  
    empName VARCHAR2(30) NOT NULL,  
    department VARCHAR2(30) NOT NULL,  
    email VARCHAR2(30) NOT NULL,  
    phone VARCHAR2(30)  
);
```

Below the query editor, the 'Query Result' and 'Script Output' tabs are visible. The 'Script Output' tab shows the execution results, indicating that the table was created successfully.

Error starting at line : 1 in command -
CREATE TABLE employee (
 empNo VARCHAR2(8) PRIMARY KEY,
 empName VAARCHAR2(30) NOT NULL,
 department VARCHAR2(30) NOT NULL,
 email VARCHAR2(30) NOT NULL,
 phone VARCHAR2(30)
)
Error report -
ORA-00907: missing right parenthesis
00907. 00000 - "missing right parenthesis"
*Cause:
*Action:

Table EMPLOYEE created.

Q5)

SQL Query:

```
INSERT INTO employee (empNo, empName, department, email, phone)
```

```
VALUES ('E001', 'Muhammad Abu Bakar', 'IT', 'bakar@gmail.com', '0300-1234567');
```

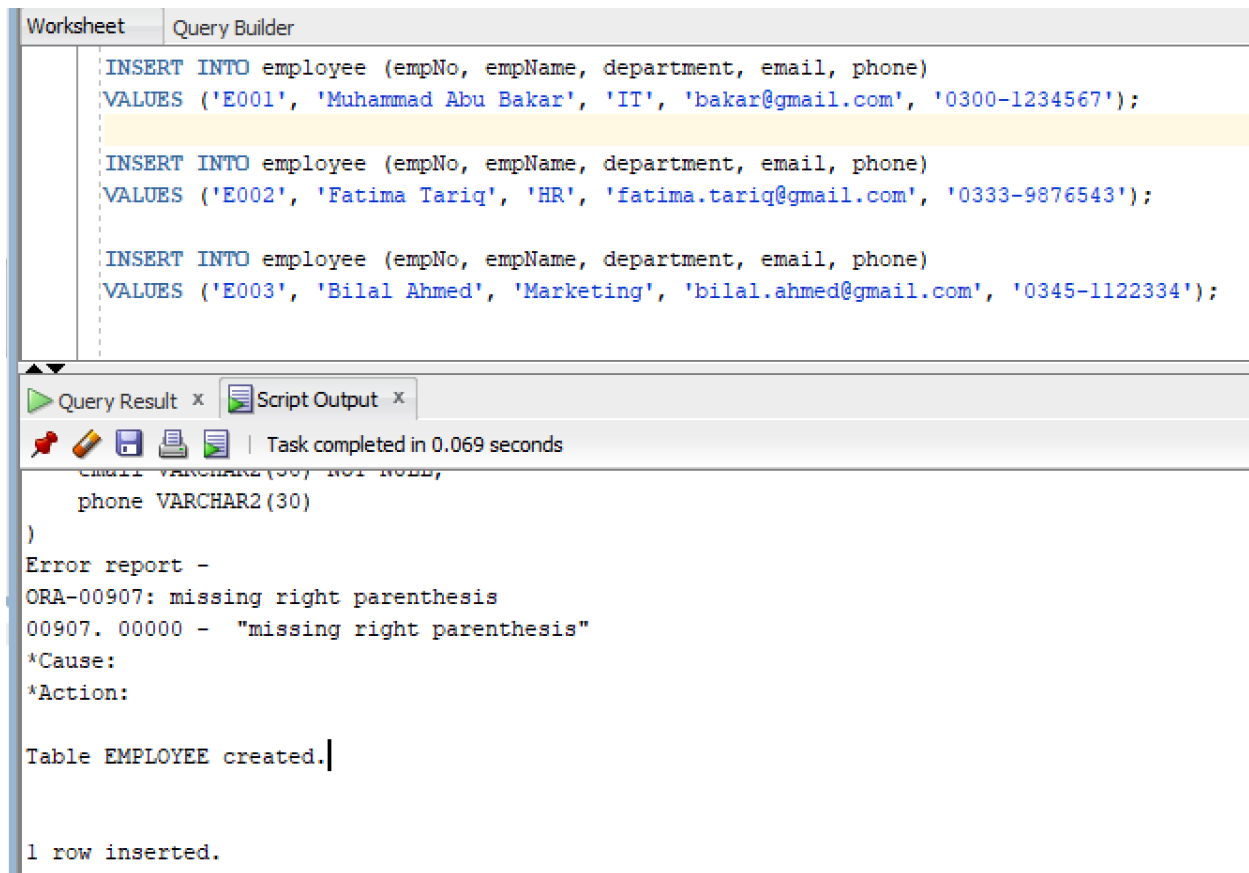
```
INSERT INTO employee (empNo, empName, department, email, phone)
```

```
VALUES ('E002', 'Fatima Tariq', 'HR', 'fatima.tariq@gmail.com', '0333-9876543');
```

```
INSERT INTO employee (empNo, empName, department, email, phone)
```

```
VALUES ('E003', 'Bilal Ahmed', 'Marketing', 'bilal.ahmed@gmail.com', '0345-1122334');
```

Snapshot of Query:



The screenshot shows a database query tool interface. At the top, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying three SQL INSERT statements. Below the queries, there is a 'Query Result' tab showing the results of the queries. The results include the table structure for 'EMPLOYEE' and the output of the queries.

```
Worksheet Query Builder

INSERT INTO employee (empNo, empName, department, email, phone)
VALUES ('E001', 'Muhammad Abu Bakar', 'IT', 'bakar@gmail.com', '0300-1234567');

INSERT INTO employee (empNo, empName, department, email, phone)
VALUES ('E002', 'Fatima Tariq', 'HR', 'fatima.tariq@gmail.com', '0333-9876543');

INSERT INTO employee (empNo, empName, department, email, phone)
VALUES ('E003', 'Bilal Ahmed', 'Marketing', 'bilal.ahmed@gmail.com', '0345-1122334');
```

Query Result x Script Output x

Task completed in 0.069 seconds

```
email VARCHAR2(50) NOT NULL,
phone VARCHAR2(30)
)
Error report -
ORA-00907: missing right parenthesis
00907. 00000 - "missing right parenthesis"
*Cause:
*Action:

Table EMPLOYEE created.

1 row inserted.
```

Q6)

1)

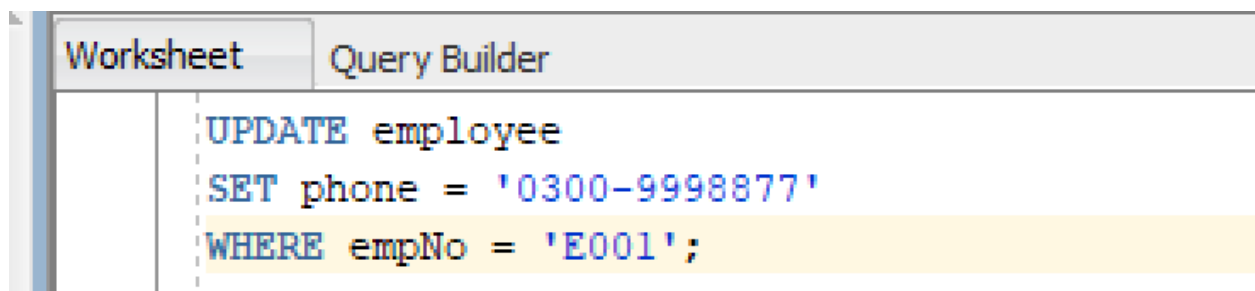
SQL Query:

UPDATE employee

SET phone = '0300-9998877'

WHERE empNo = 'E001';

Snapshot of Query:



1 row updated.

2)

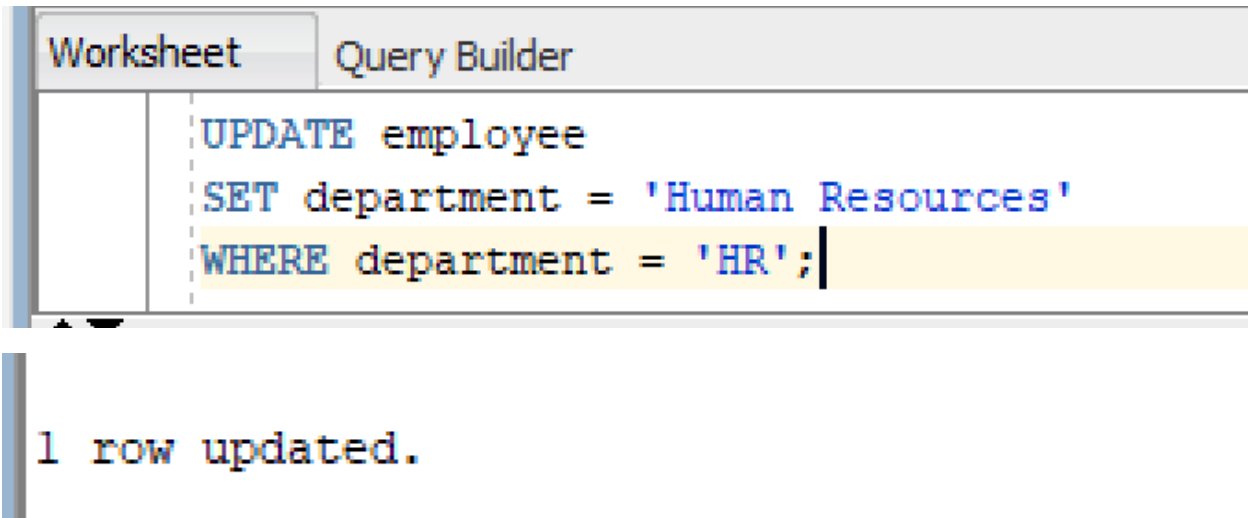
SQL Query:

UPDATE employee

SET department = 'Human Resources'

WHERE department = 'HR';

Snapshot of Query:



3)

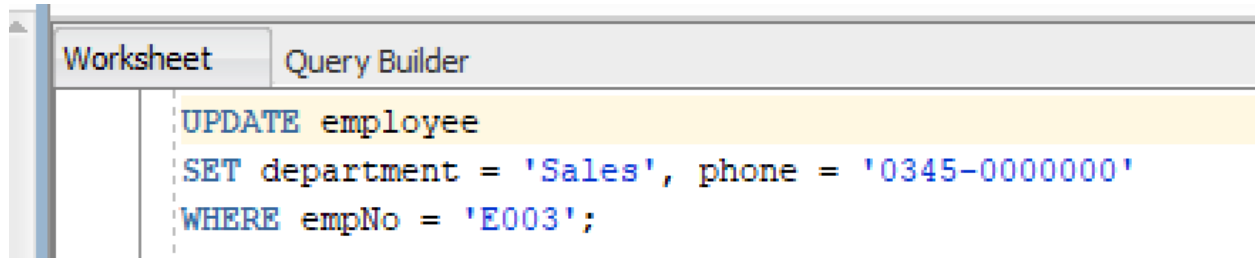
SQL Query:

UPDATE employee

SET department = 'Sales', phone = '0345-0000000'

WHERE empNo = 'E003';

Snapshot of Query:



1 row updated.

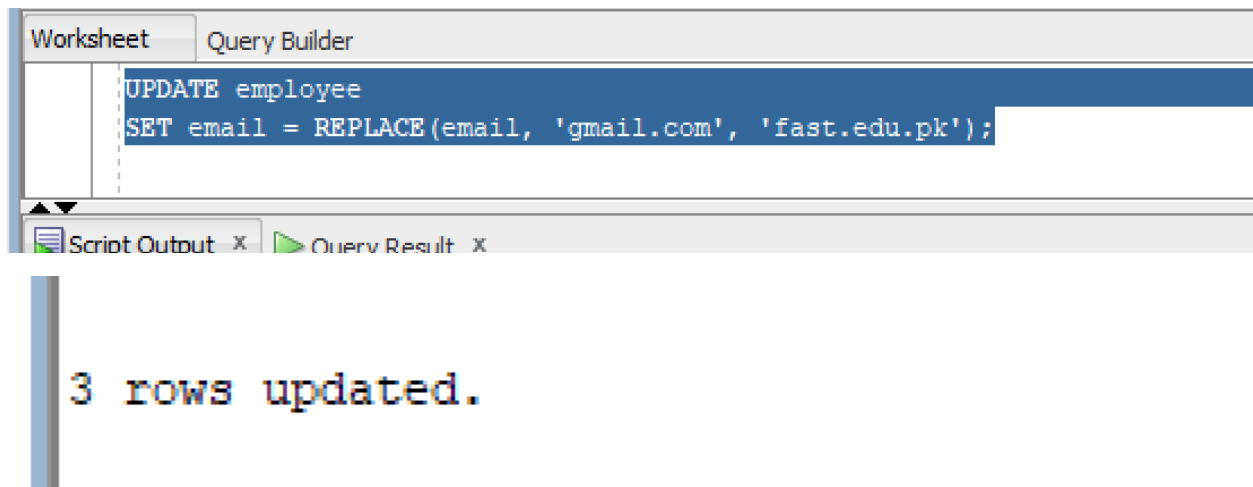
4)

SQL Query:

```
UPDATE employee
```

```
SET email = REPLACE(email, 'gmail.com', 'fast.edu.pk');
```

Snapshot of Query:



Q7)

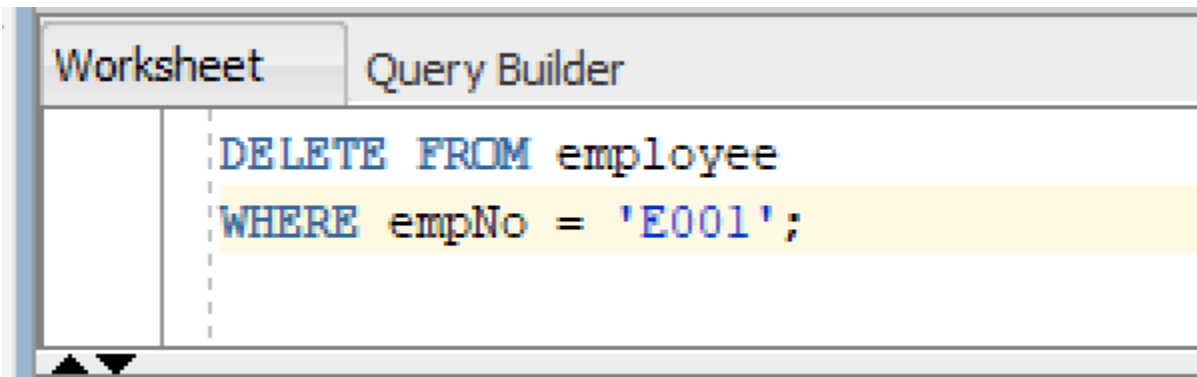
1)

SQL Query:

DELETE FROM employee

WHERE empNo = 'E001';

Snapshot of Query:



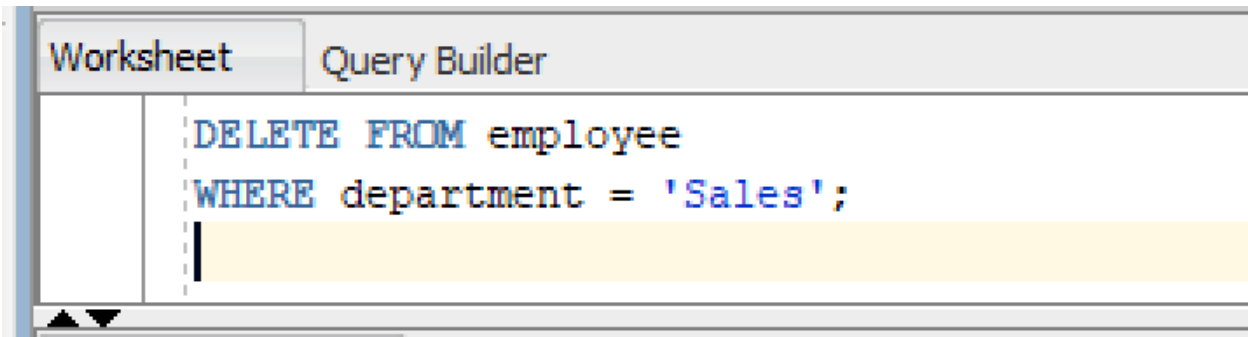
1 row deleted.

2)

SQL Query:

```
DELETE FROM employee  
WHERE department = 'Sales';
```

Snapshot of Query:



3)

SQL Query:

```
DELETE FROM employee;
```

Snapshot of Query:

Worksheet Query Builder

DELETE FROM employee;

Script Output x Query Result x

     | Task completed in 0.05 seconds

Error starting at line : 7 in command -
INSERT INTO employee (empNo, empName, department, email, phone)
VALUES ('E003', 'Bilal Ahmed', 'Marketing', 'bilal.ahmed@gmail.com', '0345-1122334')
Error report -
ORA-00001: unique constraint (SYS.SYS_C007215) violated

1 row inserted.

1 row inserted.

1 row updated.

1 row updated.

1 row updated.

3 rows updated.

1 row deleted.

2 rows deleted.

4)

SQL Query:

TRUNCATE TABLE tasks;

Snapshot of Query:

